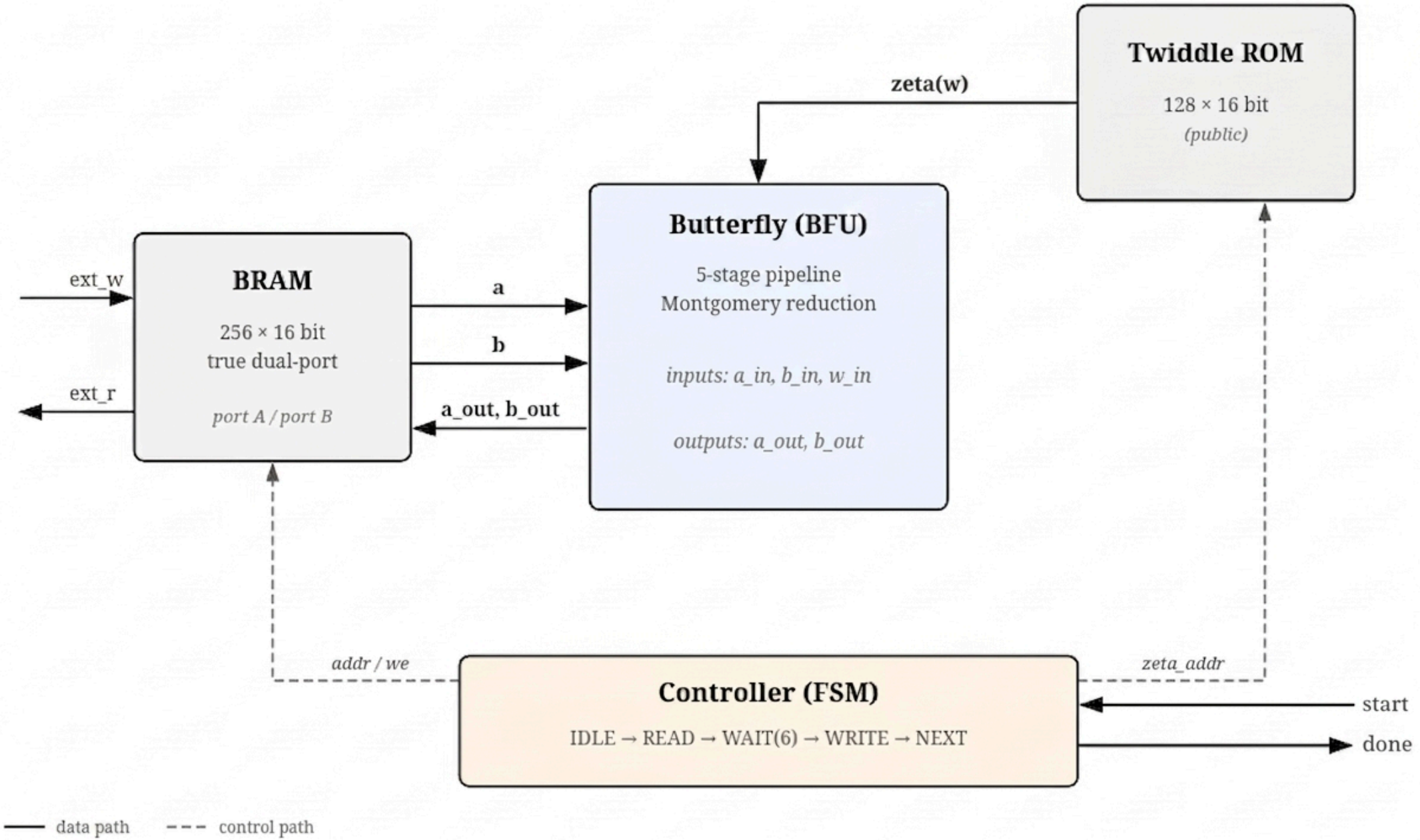

Twiddle ROM
128 × 16 bit
(public)
zeta(w)
Butterfly (BFU)
5-stage pipeline
Montgomery reduction
inputs: a_in, b_in, w_in
outputs: a_out, b_out
BRAM
256 × 16 bit
true dual-port
port A / port B
ext_w
ext_r
a
b
a_out, b_out
addr / we
zeta_addr
Controller (FSM)
IDLE → READ → WAIT(6) → WRITE → NEXT
start
done
data path
control path